

TOPIC 3, PART B

Elasticity and the Applications of Demand

Principles of Microeconomics

Prof. Bryan Cutsinger • Florida Atlantic University

Half the Oranges, More Dollars. Why Did Revenue Rise?

A frost kills half the orange crop. Growers take in more total dollars than the full crop would have brought.

You already know the buyers lost surplus. Today's question is different:

What must be true of the demand for oranges for revenue to rise when the quantity falls?

Last Time: Which Way. Today: How Much.

WRAPPED UP

Marginal worth, the buyer's surplus, and what a price really is

The demand curve is built.

TODAY

How big is the response, and who cares?

Elasticity, revenue, time, and the real price.

TUESDAY

Why do both sides of a trade gain?

Gains from exchange.

Most of today is problems, again.

01

Measuring the Response

Elasticity is a ratio of two percentages, and it is never the slope.

Elasticity Is a Ratio of Two Percentages

Price up 1%. Quantity demanded down 3%.

(\$10 to \$10.10 is a 1% rise.)

Price elasticity of demand the percentage change in quantity demanded divided by the percentage change in price, reported as a nonnegative magnitude: $e = |\% \Delta Q / \% \Delta P|$.

TEST IT Price falls 1%, quantity rises 3%. Still 3?

Elasticity Is a Ratio of Two Percentages

Price up 1%. Quantity demanded down 3%. $e = 3$
($\$10$ to $\$10.10$ is a 1% rise.)

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TEST IT Price falls 1%, quantity rises 3%. Still 3?

Elastic, $e > 1$: quantity responds more than proportionally.

Inelastic, $e < 1$: less than proportionally — not zero.

Measure Each Change Against the Midpoint

\$10 down to \$9 is a 10% cut. \$9 back up to \$10 is an 11.1% rise. Same step — which percentage?

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THE CONVENTION

$$\% \Delta Q = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \times 100$$

$$\% \Delta P = \frac{P_2 - P_1}{(P_1 + P_2)/2} \times 100$$

$$e = | \% \Delta Q / \% \Delta P |$$

The midpoint is the average of the two endpoints, so the answer no longer depends on the direction of travel.

Measure Each Change Against the Midpoint

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$$e = | \% \Delta Q / \% \Delta P |$$

WORKED: \$6 → \$5, QUANTITY 5 → 6

$$\% \Delta Q = \frac{1}{5.5} \times 100 = 18.2\%$$

$$\% \Delta P = \frac{-1}{5.5} \times 100 = -18.2\%$$

$$e = |18.2 / -18.2| = 1: \text{unit elastic}$$

The midpoint is the average of the two endpoints, so the answer no longer depends on the direction of travel.

Elasticity Changes Along the Same Straight Line

Price	Quantity
\$10	1
9	2
8	3
7	4
6	5
5	6
4	7
3	8
2	9
1	10

One straight-line demand schedule: every row is one point on the same curve.

Elasticity Changes Along the Same Straight Line

Price	Quantity	Market value
\$10	1	\$10
9	2	18
8	3	24
7	4	28
6	5	30
5	6	30
4	7	28
3	8	24
2	9	18
1	10	10

Market value is price times quantity. Watch it rise, flatten at \$30, and fall — along one unchanged curve.

Elasticity Changes Along the Same Straight Line

Price	Quantity	Market value	% price cut	% quantity rise
\$10	1	\$10		
9	2	18		
8	3	24		
7	4	28		
6	5	30		
5	6	30	18.2%	18.2%
4	7	28		
3	8	24		
2	9	18		
1	10	10		

Each row's right-hand entries compare that row with the row above, on midpoint bases.

The bold row is the step you just computed: 18.2% against 18.2%, $e = 1$. Every other row is the same arithmetic on a different pair.

Elasticity Changes Along the Same Straight Line

Price	Quantity	Market value	% price cut	% quantity rise
\$10	1	\$10		
9	2	18	10.5%	66.7%
8	3	24	11.8%	40.0%
7	4	28	13.3%	28.6%
6	5	30	15.4%	22.2%
5	6	30	18.2%	18.2%
4	7	28	22.2%	15.4%
3	8	24	28.6%	13.3%
2	9	18	40.0%	11.8%
1	10	10	66.7%	10.5%

Each row's right-hand entries compare that row with the row above, on midpoint bases.

The same \$1 cut is a small percentage of a high price and a large one of a low price; one more unit is a large share of a small quantity and a small share of a large one.

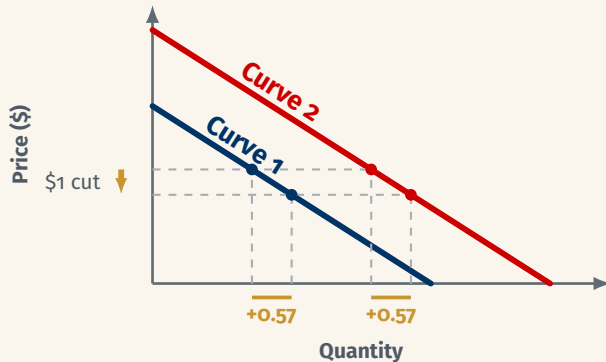
Elasticity Changes Along the Same Straight Line

Price	Quantity	Market value	% price cut	% quantity rise	Elasticity
\$10	1	\$10			
9	2	18	10.5%	66.7%	6.3
8	3	24	11.8%	40.0%	3.4
7	4	28	13.3%	28.6%	2.1
6	5	30	15.4%	22.2%	1.4
5	6	30	18.2%	18.2%	1.0
4	7	28	22.2%	15.4%	0.69
3	8	24	28.6%	13.3%	0.47
2	9	18	40.0%	11.8%	0.29
1	10	10	66.7%	10.5%	0.16

Each row's right-hand entries compare that row with the row above, on midpoint bases.

Elasticity falls as you move down the line: 6.3 on the top step, 1.0 on the \$6-to-\$5 step that straddles the peak, and 0.16 on the bottom step. State the price interval.

Parallel Curves, Same Step, Different Elasticities

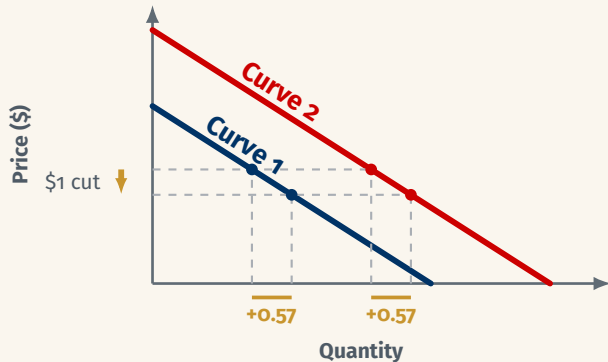


THE SAME \$1 STEP, \$4.50 $\rightarrow$ \$3.50

	Curve 1	Curve 2
Quantity change	?	?
% price cut	?	?
% quantity rise	?	?
Elasticity	?	?

Two exactly parallel demand curves, and one \$1 price cut applied to both. Which has the larger arc elasticity over this interval?

Parallel Curves, Same Step, Different Elasticities



THE SAME \$1 STEP, \$4.50 $\rightarrow$ \$3.50

	Curve 1	Curve 2
Quantity change	+0.57	+0.57
% price cut	25%	25%
% quantity rise	33%	17%
Elasticity	1.33	0.67

Same slope, same quantity change, half the elasticity: curve 2 measures the same 0.57 units against a bigger quantity. Steepness is not elasticity.

02

Price and the Seller's Revenue

Whether a price cut brings in more money depends on where you sit on the curve.

Whether a Price Cut Helps a Seller Depends on Elasticity

FROM THE TABLE

Price	Quantity	Value
\$7	4	\$28
6	5	30
5	6	30
4	7	28
3	8	24

Cutting the price above the bold row raised the take.

Below it, the same cut lowered it. **Why?**

FILL IN THE GRID: TOTAL REVENUE

	Elastic ($e > 1$)	Unit ($e = 1$)	Inelastic ($e < 1$)
Price rises	?	?	?
Price falls	?	?	?

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FILL IN THE GRID: TOTAL REVENUE

	Elastic ($e > 1$)	Unit ($e = 1$)	Inelastic ($e < 1$)
Price rises	Revenue falls	Unchanged	Revenue rises
Price falls	Revenue rises	Unchanged	Revenue falls

Elastic: quantity wins. Inelastic: price wins.

Revenue is not profit, and neither one says buyers are better off.

Half the Crop, More Dollars. What Does That Say About Buyers?

Run the frost on our schedule. Which pair of rows shows a crop cut in half with the growers taking in more?

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Run the frost on our schedule. Which pair of rows shows a crop cut in half with the growers taking in more?

Halve the crop from...	Quantity	Price	Revenue	Elasticity
the \$3 row	8 → 4	\$3 → \$7	\$24 → \$28	0.83 : inelastic

Half the Crop, More Dollars. What Does That Say About Buyers?

Run the frost on our schedule. Which pair of rows shows a crop cut in half with the growers taking in more?

Halve the crop from...	Quantity	Price	Revenue	Elasticity
the \$3 row	8 → 4	\$3 → \$7	\$24 → \$28	0.83 : inelastic
the \$5 row	6 → 3	\$5 → \$8	\$30 → \$24	1.44 : elastic

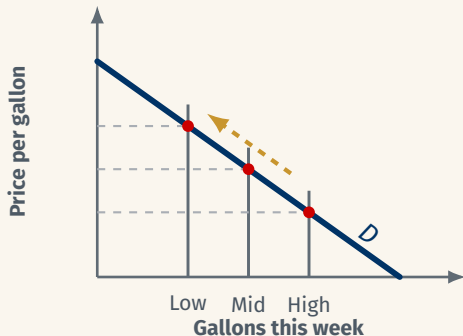
Same event, opposite answer. The only difference is where on the curve it happened.
Rising revenue after a quantity cut is the signature of inelastic demand over that interval.

Less Gas Slides Up the Curve. More Drivers Shift the Curve.

Gas prices jumped this weekend. A slide along the curve, or a new curve?

Less Gas Slides Up the Curve. More Drivers Shift the Curve.

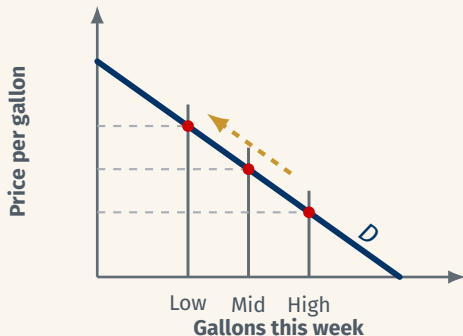
STOCKS CHANGE, DEMAND DOES NOT



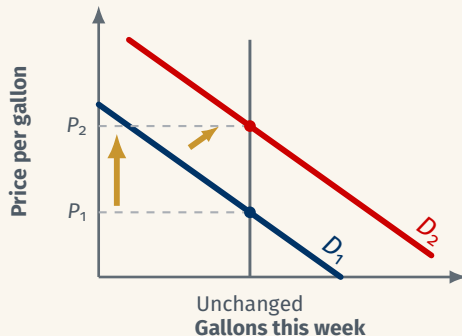
The week's stock is fixed at a low, mid, or high level. Less gas on hand selects a higher price on the same curve: a slide, and elasticity describes it.

Less Gas Slides Up the Curve. More Drivers Shift the Curve.

STOCKS CHANGE, DEMAND DOES NOT

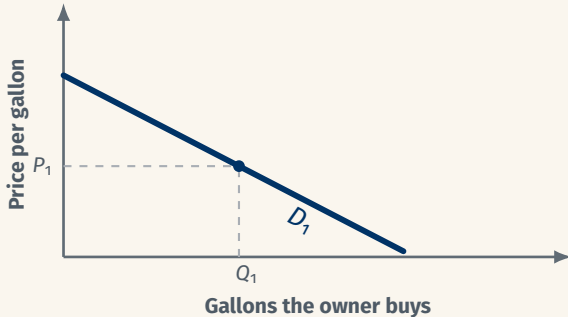


DEMAND CHANGES, STOCKS DO NOT



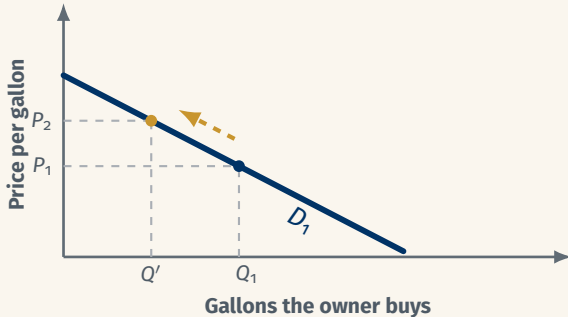
A holiday puts more drivers on the road: the whole curve shifts, the price jumps at the same quantity, and no elasticity of demand describes that.

The Gas-Station Owner: Which Effect Wins?



Gas sells at P_1 ; the owner, as a driver, buys Q_1 . Then the price of gas rises to P_2 .

The Gas-Station Owner: Which Effect Wins?



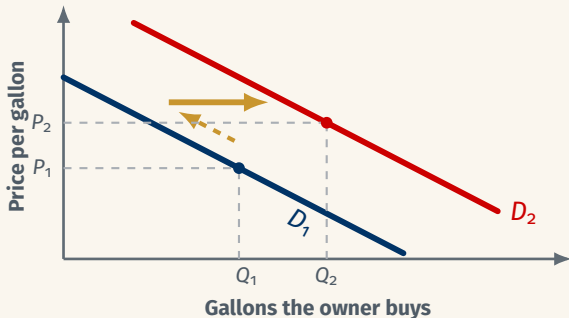
SUBSTITUTION EFFECT

Dearer gas: lean away

Always toward less: a slide up D_1 .

Substitution alone slides the owner up D_1 to Q' : fewer gallons.

The Gas-Station Owner: Which Effect Wins?



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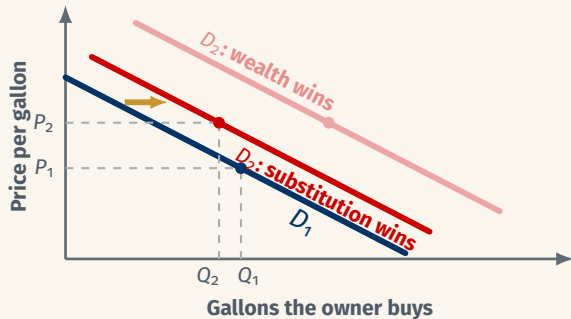
WEALTH EFFECT

Owens the gas he sells

A net seller of gasoline gets richer, so the curve shifts right.

Wealth effect wins: the shift is large and the owner buys more, $Q_2 > Q_1$. Both curves still slope down.

The Gas-Station Owner: Which Effect Wins?



SUBSTITUTION EFFECT

Dearer gas: lean away

Always toward less: a slide up D_1 .

WEALTH EFFECT

Owens the gas he sells

A net seller of gasoline gets richer, so the curve shifts right.

Substitution effect wins: the shift is small and the owner buys less, $Q_2 < Q_1$. Only the size of the shift differs.

03

Time

The response to a price change keeps growing long after the price moved.

“A 10% Gas Price Rise Won’t Change My Driving.” Law Broken?

Gas is up 10%. Next week you drive to campus exactly as before. Where, and when, would a response show up?

“A 10% Gas Price Rise Won’t Change My Driving.” Law Broken?

Gas is up 10%. Next week you drive to campus exactly as before. Where, and when, would a response show up?

THIS WEEK

Drive a little less

Combine errands, skip the optional trip. The cheapest margins move first — for some people, not all.

THIS YEAR

Carpool, ride the bus, sell the truck

Arrangements change once the price looks permanent.

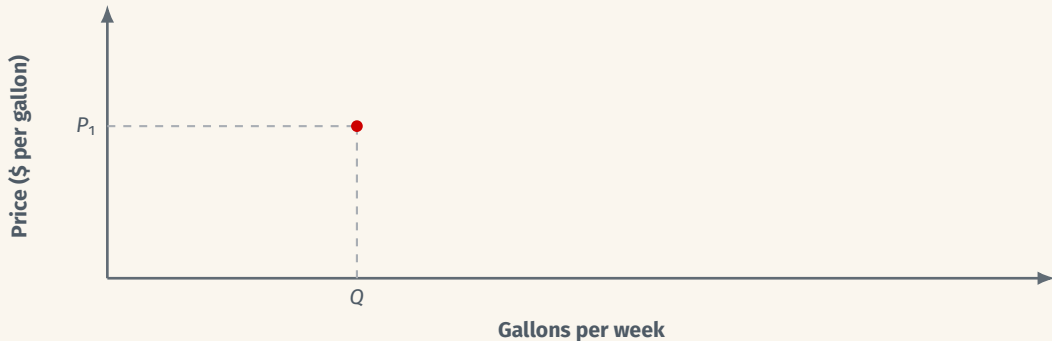
THREE YEARS

Smaller cars, shorter commutes, new fleets

Old equipment wears out; what replaces it is built for the new price.

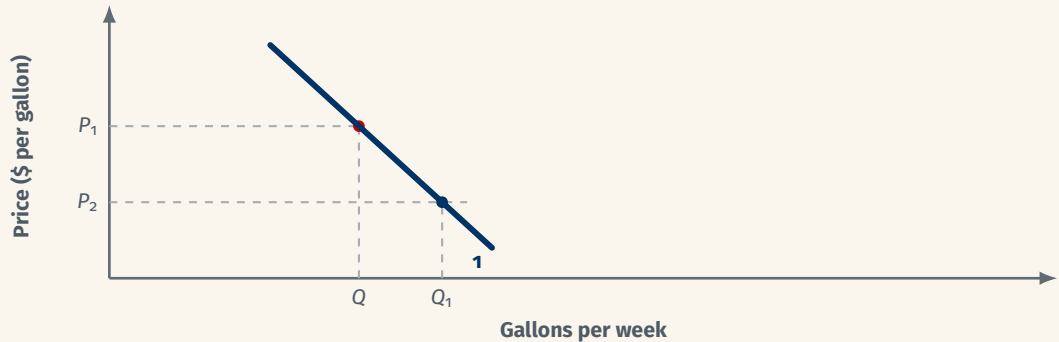
Not broken. The law never promised an instant or universal response. Give it time.

The Longer a Price Change Has Held, the Bigger the Response



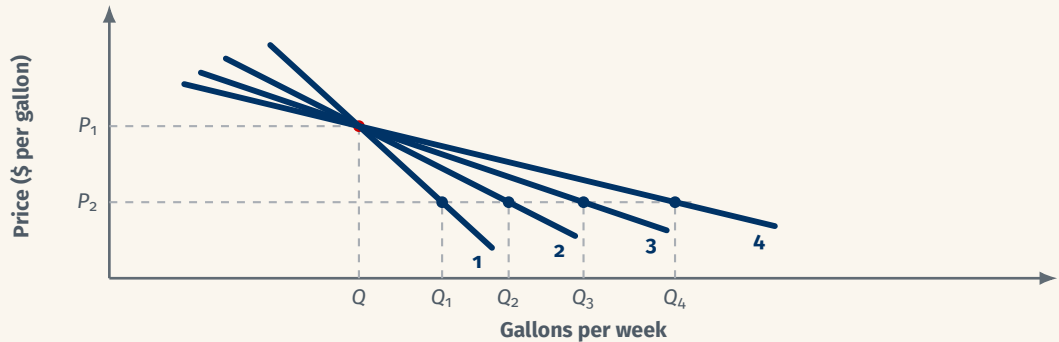
Gasoline sells at P_1 ; drivers buy Q . Now the price falls and stays down.

The Longer a Price Change Has Held, the Bigger the Response



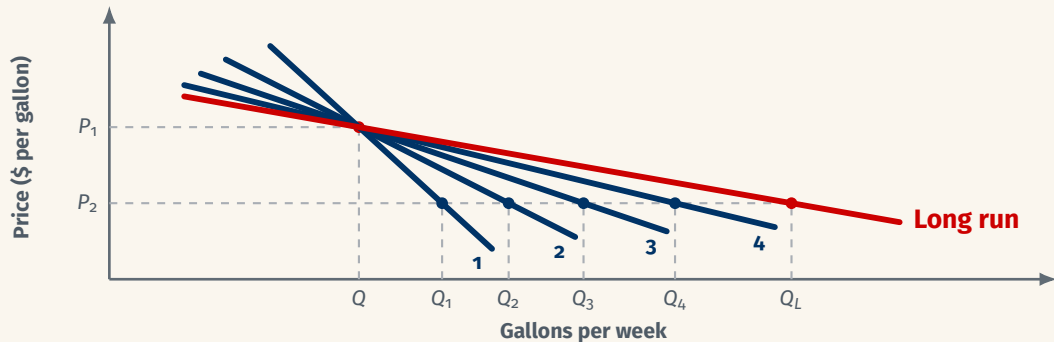
In the first weeks only curve 1 applies: same cars, same commutes, so quantity moves just to Q_1 .

The Longer a Price Change Has Held, the Bigger the Response



As months pass the operative curve flattens through 2, 3, and 4, and purchases at the same P_2 climb.

The Longer a Price Change Has Held, the Bigger the Response



Same starting point, same price cut, so the bigger quantity response is the more elastic curve — here also the flatter one. Demand grows more elastic with time.

04

The Real Price Is the Relative Price

What you give up in other goods, not the dollar figure, governs the choice.

Candy Costs More Dollars. Did It Get More Expensive?

Candy goes from \$4 to \$5 a pound. Ice cream goes from \$2 to \$3 a quart. Did candy get more expensive?

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	Before	After	Change
Candy, per pound	\$4	\$5	+25%
Ice cream, per quart	\$2	\$3	+50%
One pound of candy costs	2 quarts	1.67 quarts	cheaper

In dollars, yes. In ice cream, candy got cheaper.

Why Is So Much of Maine's Best Lobster Eaten in Chicago?

The regions that grow the finest produce ship it away and keep the ordinary grade. Why?

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The regions that grow the finest produce ship it away and keep the ordinary grade. Why?

	California price	+ shipping	= New York price
Choice grapes	\$1.00	\$0.50	\$1.50
Standard grapes	\$0.50	\$0.50	\$1.00
One pound of choice costs	2 standard	→	1.5 standard

Add the same charge to both grades and the better grade gets relatively cheaper wherever it lands.

Four Tools Turn “How Much” Into a Prediction

- 01 Elasticity: two percentages, midpoint bases, reported for a price interval. Never the slope.
- 02 Revenue: elastic, quantity wins; inelastic, price wins. Read the frost backwards.
- 03 Time: the response keeps growing after the price moves. Put a clock on each margin.
- 04 Relative price: ask “relative to what?” A fixed charge favors the dear grade.

TUESDAY Both sides of a trade walk away better off. How can that be, when nothing new was made?

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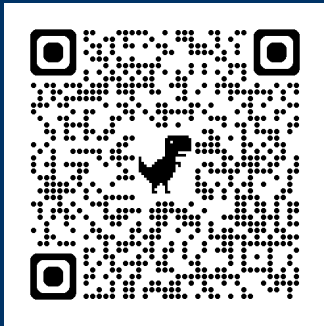
A toll road raises the toll and collects less. Which tool, and what does it tell you?

A flat \$8 delivery fee lands on a \$40 coffee order and a \$12 one. Which gets relatively cheaper?

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BEFORE YOU GO

Tell me what worked and what didn't.



Scan for a one-minute anonymous form.